Higher National Diploma in Engineering
(Civil/Electrical/Mechanical)

Second Year, Second Semester Examination – 2021

IT2203 Information Technology II B

Answer any 4 questions **Time: 2 hours**

Question 01 (Total 25 Marks)

ADO Data Control (ADODC) can be used to connect a VB Application and a Database.

i. Write down the steps to add Microsoft ActiveX Data Objects 6.0 Library to a VB project. (03 Marks)

ii. Write down the steps to add ADODC to the VB6 toolbox. (03 Marks)

iii. Draw the appearance of the ADODC in VB6 form and name the components. (06 Marks)

iv. Name the properties of ADODC that use to

a. Make a connection with a Database (02 Marks)
b. Get data from a table (02 Marks)

v. Assume that you are going to connect a MS Access database (PartsDB.mdb which is saved in C:\Myfiles) with your VB application. Write down the values to be assigned to the following properties

a. Provider (03 Marks)
b. Data source (03 Marks)

vi. After creating the connection with the database, assume that you are going to get data from the table **PartsInfo** of the database **PartsDB.mdb**. What would be the value of RecordSource property if you get all the fields of that table? (03 Marks)

Question 02 (Total 25 Marks)

i. Using examples, explain the difference between data and information. (03 Marks)

ii. What is a database? Give two examples for DBMSs that you can use to create a database. (03 Marks)

iii. Using a suitable example briefly explain the term **Primary Key**. (04 Marks)

Consider the *StuResults* table given below to answer the questions iv, v and vi

stulD	**subjectID**	**Grade**	**Marks**
S001	EN1212	A	78
S001	IT2203	B	66
S002	EN1212	C	58

B

B

iv. Identify A and B. (02 Marks)

v. Mention the most suitable data types for the fields stulD, subjectID, Grade and Marks. (04 Marks)

vi. Write SQL statements to display the following outputs

a. To display all data fields of the table StuResult (03 Marks)

b. To display stulD, subjectID and Grade of all the students (03 Marks)

c. To show all student's IDs who got A grades. (03 Marks)

Question 03 (Total 25 Marks)

As an Engineering student you are asked to create a computer application using VB 6.0 to manage the data of the following table (*stuInfo*)

StuID	**SName**	**Course**	**Contact**
0001	Nimal	HNDME	0771425366
0002	Rumesh	HNDCE	0781528369
0003	Raja	HNDQS	0714968235

i. Design a VB interface to display all data fields of the above table and write down names given to all the controls that you have used. (06 Marks)

ii. If you use ***DAO method*** to connect above created interface with the database, mention the ***reference(s)*** that you need to add before start coding. (04 Marks)

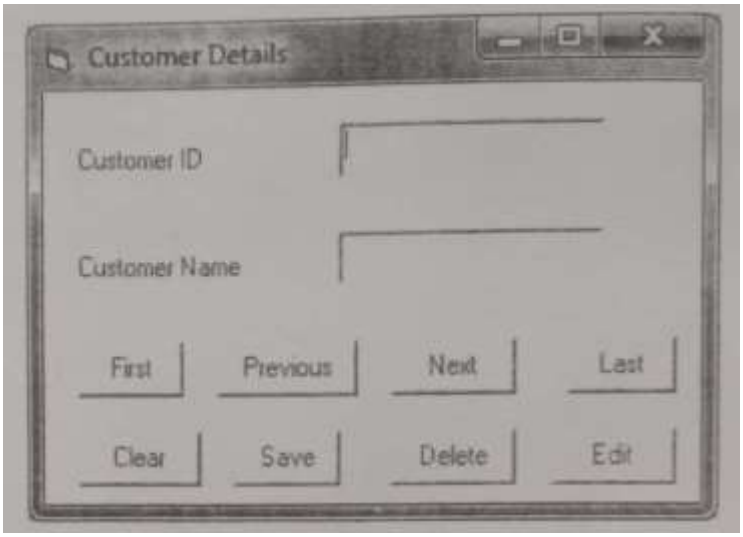
iii. Suppose your database (***stuDB.mdb***) is stored in the location ***D:\myfiles***. Write codes to create a connection with the database. (04 Marks)

iv. Write down codes to create a record set and hold **all fields** of the table stuInfo. (05 Marks)

v. Write codes to show the data of the first record in the text boxes of the interface you created above. (06 Marks)

Question 04 (Total 25 Marks)

Consider that you have created the following VB interface and connected it with the database **CustomerDB.mdb** using DAO method.



The variable **RSI** (Recordset) has been created and filled with data of CustomerInfo table (CusId and CusName are the fields of **CustomerInfo** table and CusId is the primary key).
Names of the text boxes: txtCid, txtCname
Names of the command buttons: cmdFirst, cmdPre, cmdNext, cmdLast, cmdClear, cmdSave, cmdDel, cmdEdit

i. Using an example briefly describe the following properties relevant to a Recordset. (03 Marks)

a. BOF
b. EOF

ii. Write codes to navigate through the records using relevant command buttons.

a. Go to the first record (02 Marks)
b. Go to the last record (02 Marks)

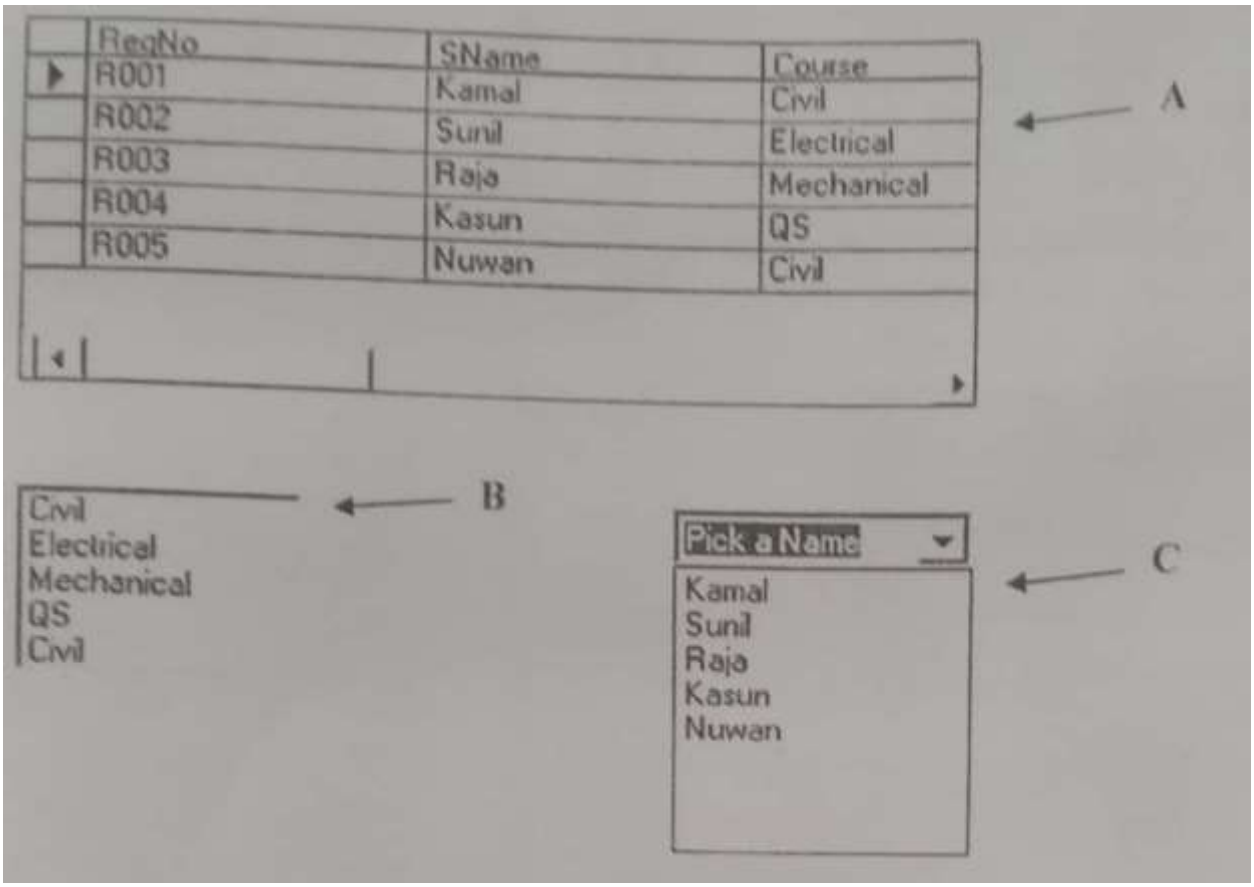
c. Go to the previous record (04 Marks)
d. Go to the next record (04 Marks)

iii. Write codes to

a. Clear the text boxes (02 Marks)
b. Add a new record to the table CustomerInfo (04 Marks)
c. Delete the current record (02 Marks)
d. Edit/ update the current record (02 Marks)

Question 05 (Total 25 Marks)

Consider the VB interface given below and answer the questions:



i. Identify data controls used in the above interface. (06 Marks)

ii. Assume that you have added an ADODC (***dataStudent***) to the interface and connection string and record source are configured to get all fields from the table ***stuInfo***. stuInfo table has three fields RegNo, SName and Course. Write down the relevant property and the value for each data control to show the output as shown in the above interface. (06 Marks)

iii. Write an event procedure (sub procedure) to make a connection with the database when it calls. (06 Marks)

Hint: name and the path of the database is D:\Myfiles\stuDB.mdb

iv. Using DAO method write codes to show all values of the field *Course* on the data control "B" As shown in the above interface. (07 Marks)